

Problem Set 10

Murphy Chesley

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Table 1: Model Comparison: Best Hyperparameters and Test Accuracy

Algorithm	Hyperparameter(s)	CV Accuracy	Test Accuracy
Logit	$\lambda = 0.00$	0.85	0.85
Tree	cost complexity = 0.00, depth = 15, min $n = 20$	0.87	0.87
NNet	hidden units = 2, $\lambda = 1.00$	0.85	0.85
KNN	neighbors = 29	0.84	0.84
SVM	cost = 1.00, $\sigma = 0.25$	0.86	0.86

The out of sample performance is roughly the same only varying by 0.03 from the best (tree) and worst (kNN). This shows that each algorithm does well out of sample and that the data is generally well estimated. No algorithm is inherently better in this case as there is not one with a drastic difference.